Hiding the peak, not missing the days: support-based screening of incomplete river-temperature archives

AUTHOR ONE[1]

[1]AFFILIATION PENDING AUTHOR CONFIRMATION

Key points

- A gap of fixed length damages the hottest 30-day mean only when it hides that window; the annual mean reacts to length
- At a 5% error cap, a rule that screens hidden peak support keeps 67% of an external archive; a gap-length rule keeps 40%
- A fitted score's 7–8 point gain over that rule needs donor and residual information, not the support share; most networks need no screen

Abstract

Incomplete river-temperature records are admitted by counting missing days: a year passes if its longest gap is short enough. A gap of a given length does not remove the same amount of the statistic computed, and in real calendars the two vary together. We pair 5,114 complete station-years from 146 river networks with 53,721 designed outages, with 1,899 more that cross gap length and position within the same station-years, and with 418 real monitoring calendars, and score admission of the annual hottest 30-day mean. Position, not length, orders the error: a 30-day gap over the peak window costs 0.64 °C more than the same gap outside it, while the annual mean responds to length. The error splits exactly into the error that imputed days inside the true window carry and a false-peak term; much of the apparent cost of a misplaced reconstructed window is hidden share. At a 5% cap on the error of the admitted reconstruction, a peak-window rule keeps 67% of a 13,017-case external panel and a gap-length rule 40%. A fitted risk score keeps 74%, but the same model without the peak-window share keeps 73%: its gain over the rule comes from donor and reconstruction diagnostics. On 418 real calendars the unscreened error is 2.6–4.1%, concentrated in two to four of 14 networks, and screens refuse hot years no more often than other records. Support orders the error; a screen's value depends on how consequential the gaps are.

Plain Language Summary

Rivers are warming, and managers ask how hot a river got in its hottest month. Answering needs a year of daily temperatures; records have holes; and the usual screening test asks how many days are missing. We show that this is the wrong question for a summer peak. What matters is whether the missing days include the few weeks the peak depends on. Using complete years from 146 river networks, we place equal gaps in different positions: a gap over the peak window spoils the hottest 30-day mean while the same gap in autumn barely moves it, and a whole-year average behaves the other way. Screening on this hidden share keeps two thirds of a test archive where a gap-length rule keeps two fifths. A learned risk score keeps slightly more, but the extra comes from knowing how well neighbouring stations can fill the hole, not from the peak window. On the holes real monitoring programmes usually leave, the peak is mostly observed, so most networks need no screen, while the few that do fail badly. A screen refuses records in proportion, hot years included, so it pays only where gaps are long and badly placed.

1 Introduction

Cold-water habitat assessment, thermal-regime classification and criterion-exceedance accounting (Washington State Department of Ecology, 2024) all reduce a year of daily water temperature to one number — a hottest 30-day mean, an annual mean, a seasonal amplitude (Caissie, 2006; Isaak et al., 2017) — and it is that number, not the daily series, that enters a management judgement. The error that matters is therefore the error of the derived statistic. On the designed outages placed on the complete station-years assembled here, computing the summer peak from the recorded days alone, which is what an agency does once a year passes a completeness test, carries a mean absolute error of 1.31 °C, and 24.9% of station-years are then wrong by more than 1.0 °C. An error of that size turns a thermally stressful year into an unremarkable one.

Current practice assumes that the amount of missing data decides whether the derived number may be used, and two families of rule encode that assumption (Table 1). Collection standards specify what must be present: Mauger et al. (2015), writing minimum standards for a state whose records feed regional analyses, require a minimum sampling duration of one calendar month and recommend a deployment period of 1 June to 31 August "to capture the thermal maxima"; the same standards govern which of those records enter the regional assessment (Mauger et al., 2017). Quantitative guidance constrains what may be absent: Johnson et al. (2021) report that "datasets with less than seven weeks of consecutively missing data enabled estimation of all annual signal parameters with reasonable accuracy", expanding to "approximately 20 weeks" when missing data are imputed. The gap-filling literature optimises the accuracy of the filled series rather than the admissibility of the completed year (Mohseni et al., 1998; Li et al., 2017), and treats the incomplete record as a missing-data problem rather than as a property of the derived statistic (Rubin, 1976), and the closest prior work fills incomplete summer records so that a thermal peak can be computed, applies that filling uniformly, declines no station-year, and admits July–August-only records on the climatological position of the hottest window (Beaufort et al., 2022). Large-sample thermal models optimise daily or reach-scale prediction skill (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023), and selective-prediction methods control error at a fixed rejection rate (Geifman & El-Yaniv, 2017; Denis et al., 2020), and distribution-free

risk control bounds an expectation under an exchangeability assumption (Lei et al., 2018; Angelopoulos et al., 2022; Bai et al., 2026). None of these rules measures how much of the target statistic's own support a record contains.

The assumption is plausible for some statistics and wrong for others. An annual mean is a sum over days, so removing k days removes k terms and it matters little which. A hottest 30-day mean is an extreme supported on a short window: if the outage misses that window, the days that define the statistic are still observed and it is recoverable however long the outage is, provided the filling does not manufacture a higher window elsewhere; if the outage covers the window the statistic must be inferred however short the outage is. The direction of a completeness rule's error should therefore be opposite in the two cases, and an average over a mixed archive would conceal the reversal. The literature is consistent with that reversal without having isolated it: reported timing effects on annual thermal parameters are inconsistent across regions, years and parameters (Johnson et al., 2021); reconstruction bias depends on where the target window falls (Beaufort et al., 2022); paired air–water analyses show how much of the thermal regime is inherited from the atmosphere (Hare et al., 2023); large-scale climate modes modulate river thermal regimes (Sanikhani et al., 2026; You et al., 2026); systematic, time-varying instrument bias rather than random noise dominates long-term monitoring error (Jackson et al., 2025); and statistic-aware completeness conventions do exist in climate monitoring, but they are defined by counts or seasons rather than by the statistic's support (Zhang et al., 2011). Irregular sampling and sampling artifacts likewise bias means and trends and decide which model structures remain trustworthy (Grey et al., 2024), and reconstruction frameworks have been built for regulated and data-sparse basins, including Chinese catchments and the Yangtze mainstem (Huang et al., 2023; Wang, 2025); those studies answer estimation and reconstruction questions, not the archive-admission question for a named annual extreme.

Testing the assumption requires separating two properties of a gap that real records entangle. In the real gap calendars replayed here, gap length and the share of the climatological peak window hidden correlate at $+0.71$ [0.44, 0.89] across 14 networks, and gap length correlates at -0.90 with the fraction of warm-season days that remains observed; a comparison across real gaps therefore cannot say whether a rule works because it avoids long gaps, because it avoids badly placed ones, or

because both simply track how much warm-season record the station has. A second obstacle is truth: in exactly the years where the admission decision matters, the target statistic is unknown by construction, so the benefit of a rule cannot be measured where it would be applied. A third is that evaluations report a rule's discrimination rather than the composition of the archive it leaves behind, and composition is what decides which catchments remain available for analysis.

To address these limitations we assemble 5,114 complete river station-years from 146 networks and nine national agencies and place 53,721 outages on them. These come from two designs that compare cases across a mixed archive (Section 2.3); the property separation itself rests on a further, smaller crossed panel of 1,899 outages in which every one of 211 station-years carries all nine combinations of three gap lengths and three positions. Admission rules are scored on held-out river networks, on an external panel that took no part in method selection but has since been opened, and on 418 real gap calendars transplanted from 59 incomplete station-years. Three questions organise the paper. First, which property of a gap predicts the error of a derived thermal statistic, and does the answer depend on which statistic is being computed? Second, how much of that dependence is the reconstructor's doing, and which part of the error that remains belongs to the days a gap hides and which to a false peak? Third, what does conditioning on support change for the error among retained station-years, which part of a fitted screen's added retention depends on support at all, and what does refusing a record cost the hot-year judgement that the archive is used to make? The third question decides whether the first two have practical value, and on the evidence below they have value only in a regime we can name in advance.

2 Study system, data and design

2.1 The statistic and its support

Let a station-year be a sequence $y = (y_1, \ldots, y_T)$ of daily mean water temperatures and let M be the set of missing days. The annual hottest 30-day mean is

$$H_{30}(y) = \max_{t \in \{30,\ldots,T\}} \frac{1}{30} \sum_{s=t-29}^{t} y_s, \tag{1}$$

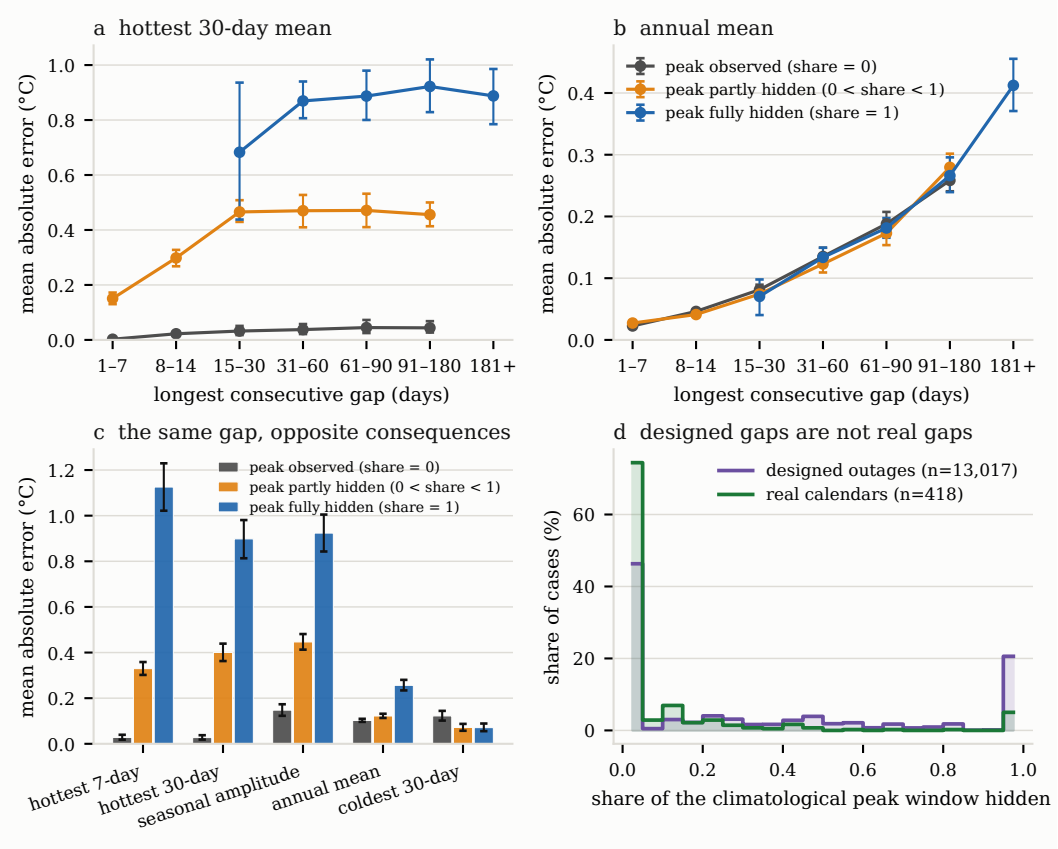


Figure 1. Hidden support, not gap length, orders the error, and the order reverses between statistics. All panels use the 13,017 designed outages of the external panel (8,040 single-run stratified and 4,977 transplanted-geometry outages) with a day-of-year climatology reconstruction, so the panels differ only in the statistic and in gap geometry. Cases are grouped by the *true* hidden share, an outcome-side variable, in a mixed archive in which length and true hidden share still correlate at +0.52; panels (a)–(c) are therefore a descriptive conditional comparison, and the controlled estimate is Figure 3a. (a) Mean absolute error of the hottest 30-day mean against the longest consecutive gap, split by the share of the true 30-day peak window that the gap hides (grey, none; orange, partial; blue, all). Errors are network-clustered means with 95% intervals (2,000 cluster-bootstrap draws over the 41 shared-station groups); a cell is drawn when it holds at least 25 cases. (b) The same for the annual mean. (c) Mean absolute error of five annual statistics at the three hidden-support classes, with the same intervals; the coldest 30-day mean moves in the opposite direction to the hottest. (d) Distribution of hidden peak support in the designed outages (purple) and in the 418 real monitoring calendars (green). Source data: `fig1_source.csv`.

Table 1. What each body of work optimises, decides on, and conditions on. The rows are ordered by what they assume about the conditioning variable, not by date.

body of work	optimises	decision unit	conditioning axis
completeness guidance (Johnson et al., 2021)	tolerable amount of missing data	station-year	outage size (timing tested; effects reported inconsistent)
collection standards (Mauger et al., 2015)	comparability across a programme	deployment	season and duration
gap filling (Mohseni et al., 1998; Li et al., 2017)	accuracy of the filled series	day	air temperature, donors
peak reconstruction (Beaufort et al., 2022)	spatial coverage of the thermal peak	reach	air regression applied uniformly; climatological summer window used to admit partial-season records
large-sample models (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023)	prediction skill	day or reach	meteorology, catchment attributes
selective prediction (Geifman & El-Yaniv, 2017; Denis et al., 2020)	error at a fixed rejection rate	case	model confidence
this work	**error of the derived statistic among retained years**	**station-year**	**share of the statistic's support hidden**

and its support is the set of days inside the winning window, $S = \{s : s \in [t^* - 29, t^*]\}$. For a gap M we distinguish two quantities that real records confound:

$$\ell = \max\{\text{length of a consecutive run in } M\}, \qquad h = \frac{|M \cap S|}{|S|}. \tag{2}$$

Here ℓ is what a completeness rule reads and h is the share of the statistic's own support that the record does not contain.

Two properties of this definition matter for what follows. First, S is defined by the year's own temperatures and is therefore unknown at the moment of the admission decision. Three windows must be kept apart. The *true* window S belongs to the complete reference year. The *climatological* window S_c is estimated from the station's other complete years; it is the only window a deployable rule can see, and the admission rules below use $\hat{h} = |M \cap S_c|/|S_c|$. The *reconstructed* window S_r is the winning window of the completed year and is an output of the reconstruction, not an input to any decision. $S_c \neq S$ is an error in the *support estimate*; $S_r \neq S$ is an error in the *reconstruction*. The diagnostic panels use the true h, and Section 4.3 separates the three rather than treating one as the cause of another. Second, h is statistic-specific: for the annual mean every day is support, so h degenerates to the missing fraction, whereas for the coldest 30-day mean the support is the winter window, and a gap that hides the summer peak is a gap that leaves the cold support observed. The same gap therefore has three different values of h and can be good for one statistic and bad for another.

2.2 Records and panels

Daily mean water temperature comes from nine national providers (ARSO, CHMI, FOEN, GKD Bayern, LUBW, NVE, Rijkswaterstaat, SYKE and USGS) and, on the external panel, from France's Hub'Eau and the UK Environment Agency. All records are public monitoring archives reused for this analysis; no sampling was conducted for this study. The base panel keeps networks with at least four complete years: 146 networks, 457 stations and 5,114 fully observed station-years. Two derived panels are used. The development panel holds 110 networks, 354 stations, 4,045 station-years and 40,704 designed outages; it was used to select methods and is not a confirmation set. The external panel holds 49 network labels, 143 distinct stations, 1,103 station-years and 13,017 designed outages. Labels that share

a station are merged into connected components before any fold is drawn, so the external panel has 41 validation groups and no target station appears in two folds; 164 station-years appear under more than one label and stay inside one group.

The stored policy reconstruction fails the 0.5 °C tolerance on 15.3% of external cases and 14.9% of development cases; those are the rates a screen has to improve on. The real, incomplete part of the archive is 2,422 station-years. Those years have no complete counterpart, so no reconstruction error can be measured on them; they are used only to describe the gap geometry that monitoring programmes actually leave.

2.3 Designed outages: three designs and what each can identify

Three designs place outages on complete station-years, and each supports a different kind of statement. The reader should not take the 53,721 outages as one factorial experiment.

Stratified outages (33,749, 62.8%) place a single contiguous run of a target length in {14, 30, 60, 90, 120, 180, 270} days so that it overlaps the *climatological* peak window S_c by none of it, a part (15–85%) or all of it. A request that no start day can satisfy is left empty: a 14-day run cannot cover a 30-day window, and a 270-day run can hardly avoid one, so three of the 21 length-by-class cells are empty or hold fewer than 100 cases in each panel. No scattered missing days are added; every stratified outage is one run. *Empirical outages* (19,972, 37.2%) transplant the lengths and calendar months of one to four gaps recorded by the same provider onto a complete year. They are the closest the designed panels come to real geometry, and they carry the entanglement of length and position with them. In the pooled panels gap length therefore still correlates with the true hidden share at +0.52 (both panels) and with the climatological overlap at +0.45 (development) and +0.46 (external), against +0.71 [0.44, 0.89] in the real calendars. Comparisons across these panels, including Figures 1 and 3c, are conditional comparisons in a mixed archive: they describe what the archive contains, not what changing a gap would do.

The controlled design is the *crossed panel*, generated separately on 24 development stations (it is not a subset of the 53,721: one third of its outages share a realisation key with a development outage and a sixth of those differ in geometry):

211 station-years from 24 stations in 12 networks, each carrying all nine combinations of a 30-, 60- or 90-day run with a climatological overlap of none, part or all, 1,899 outages in all, each reconstructed by five methods fitted leave-one-year-out. Length and climatological overlap are uncorrelated there (0.00 [−0.02, 0.03]); length and the true hidden share remain correlated at +0.14 because the true window does not coincide with S_c, so a longer run covers more of it within the same class; the position contrasts below are therefore formed within station-years and the length contrasts at a fixed true hidden share. Cells that cannot be filled are reported as empty, not replaced by other station-years. Every identification claim in Sections 4.1 to 4.3 rests on this panel; the admission results in Section 4.4 rest on the stratified and empirical outages. The five reconstruction methods see the same information on every design (Table 2).

2.4 Real calendars

The 418 candidate groups of the real-calendar replay are built by transplanting every missing day of a genuinely incomplete year onto a complete target year, keeping the calendar position of the gap. The groups come from 59 incomplete source station-years and fall on 67 target station-years in 14 networks, so the count of groups overstates the independent evidence by roughly a factor of six. This is the only setting in which the gap geometry is the one an agency actually faces. It has no new sealed panel and it is not a confirmation set; it is a development-period probe of what screening would do to real calendars.

3 Methods

3.1 Reconstruction and the information it may use

Every reconstructor is fitted leave-one-year-out: the model that fills days in year y sees only other years at the same station, plus, where the method allows it, other stations of the same network. Five reconstructors are used: day-of-year climatology, a smoothed climatology, linear interpolation between observed days, a ridge regression on visible donor stations, and a ridge regression on gridded air temperature (NASA POWER (NASA Langley Research Center, 2026)). A stored per-case policy selects one of them for each station-year on validation evidence,

Table 2. Panels, their units, and what each is allowed to support. "Groups" are connected components of networks that share a station id and are the resampling and fold unit throughout.

panel	cases	networks	groups	role
complete base	5,114 station-years	146	—	reference truth for every recovery experiment; USGS reference-network criteria follow (Falcone, 2011)
development	40,704 designed outages	110	110	method selection and cross-fitted development evidence
external	13,017 designed outages	49 labels, 143 stations	41	cross-fitted external comparison; already opened, not a new seal
crossed (mechanism)	1,899 outages on 211 station-years, 24 stations	12	12	the only controlled design: all nine length × position cells in every station-year; position and length contrasts, Figures 3a–b and 4a–c
real archive	2,422 incomplete station-years	—	—	gap geometry only; no complete counterpart, so no error can be measured
real calendar replay	418 groups from 59 source gaps on 67 target station-years	14	14	development-period probe of deployment on real gap calendars

and the reconstruction that the admission rule accepts is that policy output, whose absolute error we write e and whose consequential failure we define as $e > 0.5\,^{\circ}\mathrm{C}$. On the external panel the policy chooses donor regression for 69% of cases, smoothed climatology for 24% and interpolation for 6%; the climatology fallback is where the policy fails, at 38% against 8% for donor regression, and it is one reason the fitted score can separate records that share the same hidden support. The tolerance is fixed before any result in this paper was computed; it is roughly half the interannual standard deviation of the hottest 30-day mean within a station, so a failure is a year whose reconstructed rank would move.

3.2 The admission problem

A rule computes a score $s(x)$ from decision-time features x and admits a record when $s(x) \le \tau$. The threshold τ is selected on calibration groups only, never on the labels of the records it will judge. We compare four scores that are available before the decision and one reference arm:

- *gap-length rule*: $s = \ell$, the longest consecutive missing run.
- *peak-window rule*: $s = \hat{h}$, the share of the climatological peak window hidden by the gap.
- *certificate bound*: $s = b$, a calibrated upper bound on the reconstruction error stored by a companion certificate model; it ranks records by predicted error and carries no label from the evaluated year.
- *fitted risk score*: a quantile gradient-boosting regressor ($q = 0.9$, depth 4, 350 iterations) on 22 decision-time features that include gap geometry, visible-donor count and correlation, the reconstructor's own training residual and the disagreement among reconstructors. It is trained on e, the error of the same policy reconstruction it is used to admit, and it is cross-fitted by group: the calibration score of each outer fold is an inner out-of-fold prediction, so no outer test label can reach that fold's threshold.
- *reconstruct everything*: the reference arm, admitting every record whose policy reconstruction is defined. It is what an agency does if no screen is worth its cost, and it is the arm against which the value of screening has to be judged.

Ties are handled once, at the score definition: a threshold is an observed value of the raw score, and admission is $s \leq \tau$, so a tie block is admitted whole in calibration, in evaluation and at deployment. No jitter is added at any stage. A separate fixed-retention comparison admits the safest 70% of each fold using a deterministic case key to break ties, which is a second deployable rule and is reported as such.

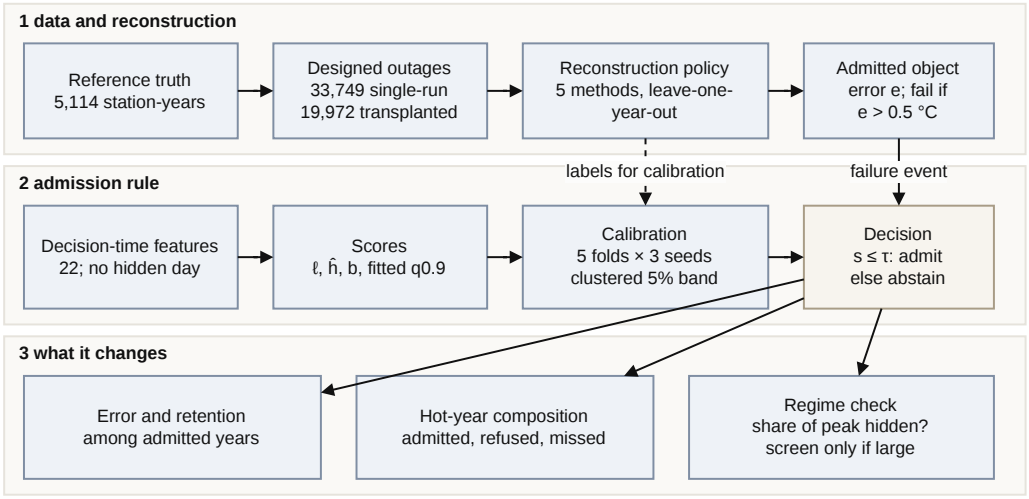


Figure 2. How a station-year becomes an admissible record. Band 1: complete station-years supply the reference truth, designed outages vary gap geometry, five reconstructors are fitted leave-one-year-out, and the stored policy output is the admitted object, whose error against the complete year defines a failure at 0.5 °C. Band 2: the admission rule reads only decision-time information, scores each record, calibrates one threshold per group fold with a clustered bound at the 5% cap, and admits a record whose score is at or below the threshold; ties are admitted whole. Band 3: what the rule changes, and the regime check that decides whether a screen is worth its cost. Schematic source: `f2_system_design.drawio` (draw.io; see the Open Research statement).

3.3 Risk control and what it means

The threshold is the largest candidate whose group-clustered bootstrap bound on the calibration false-admit rate is at most 5%, with the percentile taken at $1 - 0.05/n_{\text{cand}}$ over the n_{cand} distinct thresholds searched. We are explicit about what this is and is not. It is an empirical, finite-sample calibration rule for a clustered design, in the spirit of distribution-free risk control but without its exchangeability

guarantee (Angelopoulos et al., 2022; Bai et al., 2026). It is not a distribution-free guarantee for an unseen archive, and a bound that holds at each candidate is not a bound that holds after maximising over candidates; the Bonferroni factor is there for that reason. Section 4.5 reports what the rule actually delivers in simulation, separating the population risk of the selected threshold from the error rate observed in a finite test batch — the two are not the same event, and only the first is a property an agency can rely on.

3.4 Uncertainty and the unit of inference

The resampling unit is the shared-station group. All intervals are percentile intervals from 2,000 group-bootstrap draws unless stated otherwise, and every contrast is paired: the same resampled groups are used for both arms. Three fold seeds (101, 202, 303) are pre-declared and all are reported. Shares are fractions throughout; we write percentage points when a difference is in points. For the hot-year analysis, the case-weighted estimate is primary and a unique station-year weighting, in which each station-year carries total weight one regardless of how many outage realisations it contributes, is reported as the pre-declared sensitivity.

3.5 Attributing the error and the gain

Four analyses answer the attribution questions, each varying one thing.

An exact decomposition of the error. Let $e_t = \tilde{y}_t - y_t$ be the difference between the completed and the true year ($e_t = 0$ on observed days), $\bar{e}_W$ its mean over a 30-day window W, and $D_W = H_{30}(y) - \bar{y}_W \geq 0$ the distance of W below the true maximum. Then

$$H_{30}(\tilde{y}) - H_{30}(y) = \max_W \big(\bar{e}_W - D_W\big) = \bar{e}_S + F, \qquad F \geq 0, \tag{3}$$

because $D_S = 0$. The *hiding channel* $\bar{e}_S$ is the error the true window inherits from imputed days inside it and is exactly zero when none of S is hidden; the *false-peak channel* F is how far another window, with its own residuals added, overshoots the true one. The identity is checked on every case. Three interventions on the recorded gap residuals, holding the truth, the gap and the reconstructor fixed, then ask what drives each channel: permuting the residuals within the gap (the marginal distribution and mean are kept, persistence is destroyed; 25 permutations per case),

removing the gap mean (fluctuations and their persistence are kept), and keeping only the gap mean. These are numerical experiments on recorded residuals, not identification of a natural process.

A ladder of information at one capacity. The fitted score is re-learned with the same learner, training years, nested cross-fit and calibration as in Section 3.2, on nested subsets of its 22 features: gap and season coverage, then donor information, then reconstruction diagnostics (training residual and estimator disagreement), then station history, then the peak-window share $\hat{h}$; the same ladder is repeated with a small logistic model. The decisive contrast is the model that cannot read $\hat{h}$ against the one that can. A model that reads the *true* hidden share is not deployable and serves only to bound how much a better support estimate could add.

Refusal against a hot-year-blind null. A rule that ignores whether a year is hot and keeps a share r of all records refuses a share $1 - r$ of the hot years. We report r minus the retention of true hot years, with a group-clustered interval, and the loss $\rho N_{\mathrm{FN}} + N_{\mathrm{FP}} + c_{\mathrm{A}} N_{\mathrm{A}}$ of judging a hot year from the admitted reconstruction, where a refused record is an abstention. The break-even abstention cost c_{A}^{*} is the cost at which a screen and reconstructing everything lose the same; the screen pays only if one abstention costs less than c_{A}^{*} wrong judgements.

An archive-level rule for whether to screen. The share of an archive's records in each cell of longest gap by climatological overlap is observable before any label is seen. Weighting class-conditional failure rates of the unscreened reconstruction, estimated on designed outages only, by that composition predicts the archive's failure rate; the rule declares "no screen" when the prediction is at most the 5% cap. Nothing in it is fitted on a real calendar, so the real replay is an out-of-domain test of it.

4 Results

4.1 The same gap has opposite consequences for different statistics

4.1.1 Support orders the error, and length barely moves it

Figure 1a and b hold the gap length in view and split a mixed archive by the true hidden support, so they describe what the archive contains. For the hottest

30-day mean the support class sets the height of the curve and the length moves it much less: a gap that leaves the peak window observed costs 0.003 °C at one week and 0.04 °C at 91–180 days, while a gap that covers the window costs 0.68 °C at 15–30 days and stays near 0.9 °C from 60 days out to beyond 180. The controlled estimate comes from the crossed panel, which holds the station-year, the reconstructor and the length fixed and moves only the position: at a length of 30 days the position effect is 0.64 °C [0.55, 0.74], and at 90 days 0.89 °C [0.76, 1.04] (Figure 3a), in agreement with the 0.65 °C difference between the grouped classes at 15–30 days. Lengthening the run from 30 to 90 days at a true hidden share of at least 0.9 adds 0.12 °C [0.06, 0.18] for the same reconstructor, one seventh of the position effect (Figure 3b). For the annual mean the same split collapses: the three support classes lie on top of one another and the error rises with length from 0.02 °C at one week to 0.26 °C by 180 days, with the one class that has longer records reaching 0.41 °C.

4.1.2 A colder statistic reverses the ordering

The reversal is not confined to the annual mean. Figure 1c compares five annual statistics at the three support classes. The hottest 7-day mean behaves like the hottest 30-day mean but more steeply (0.03 against 1.13 °C). The seasonal amplitude, which is a difference of a summer and a winter window, behaves like the hot statistics because its summer term is the one at risk. The coldest 30-day mean moves the other way: 0.12 °C when the gap misses the peak window against 0.07 °C when it covers it, because a gap placed away from the summer peak is a gap that can cover the winter minimum, and a gap over the summer peak leaves the cold support observed. One rule, three statistics, opposite consequences: what a completeness count cannot express is which days a statistic is made of.

Figure 1d adds the reason this matters for practice. In the designed panel 30% of outages hide more than half of the climatological peak window and the median gap is 30 days. In the 418 real calendars the median gap is 7 days, the median missing fraction is 2.2%, and 5.7% of candidates hide more than half of that window. The two populations are not the same population, and Section 4.4 shows what that difference does to the value of a screen.

4.2 The reconstructor sets the height, not the shape

The support effect is conditional on what the reconstructor can infer. Figure 3a crosses gap position with gap length at 30, 60 and 90 days for five reconstructors, within the same station-years. Hiding the climatological peak window always costs something, but the cost ranges from 0.24 °C [0.20, 0.28] for donor-station regression to 0.64 °C [0.55, 0.74] for day-of-year climatology at 30 days. The ordering of the methods is not stable in length: linear interpolation is the second cheapest at 30 days (0.34 °C) and the most expensive at 90 days (1.31 °C), while the other four keep their order.

Figure 3b asks the complementary question: with the true hidden share held high, does length still matter? Two pre-declared cut-offs are shown, both formed within station-years. At a share of at least 0.9 (48 station-years in 12 networks) a 90-day run costs more than a 30-day run by 0.12 °C [0.06, 0.18] for climatology, 0.14 °C [0.08, 0.20] for smoothing, 0.12 °C [0.07, 0.18] for air regression and 0.05 °C [0.02, 0.08] for donor regression; at a share of at least 0.999 only 11 station-years in six networks qualify and the donor estimate (0.02 °C [−0.01, 0.05]) is the only one whose interval includes zero. These length effects are small beside the position effect at the same lengths, roughly 5 to 8 times smaller, but they are distinguishable from zero. Linear interpolation is the large exception, at 0.82 °C [0.64, 1.00] and 1.12 °C [0.75, 1.45]: when the window is entirely unobserved an interpolator must bridge the whole gap and its error grows with the bridge, whereas a climatological or donor method substitutes an estimate that does not depend on the length of the missing run. An earlier reading of the same crossed panel, using a network-paired subset estimator that needs both lengths only somewhere in the same network, put the climatology and smoothing effects at a few hundredths of a degree with intervals that include zero; that estimator discards the pairing within station-years and is reported only as a sensitivity in the supporting information.

Figure 3c puts the two findings on one axis for the four reconstructors that have a series on the external panel (that panel carries no air-temperature series). Across the full range of hidden support every reconstructor's error rises monotonically with the share hidden, but the curves are at different heights and the interpolation curve bends upward at the right-hand end. The support share is the

right conditioning variable for every method; the coefficient on it, and whether a length term survives alongside it, are properties of the reconstructor. A claim that position dominates length as a law would therefore be wrong, and it is not the claim we make.

4.3 The error splits into a hiding channel and a false-peak channel

4.3.1 Hiding and false peaks are separate channels with opposite signs

Equation 3 holds to 2×10^{-13} °C on all 9,449 finite cases of the crossed panel (46 donor or air fills are undefined), and the hiding channel is exactly zero in every case whose true window is fully observed. Figure 4a splits the error by the true hidden share. With the true window observed, the day-of-year climatology reconstruction is wrong by 0.06 °C [0.04, 0.09] on average, all of it false peak, and in 9% of those cases the false peak exceeds 0.1 °C. When the window is entirely hidden the two channels pull in opposite directions: the hidden days are filled too cool (hiding channel −0.75 °C [−0.92, −0.61]) and the filling invents a higher window elsewhere (false peak +0.43 °C [0.35, 0.52]), so the mean absolute error of 0.92 °C is larger than the net signed error. A donor regression, which carries information about the year, has the same signs at about a fifth of the size (−0.17 and +0.06 °C). The statement that a record whose peak window is observed is unaffected by a gap of any length is therefore true of the hiding channel only; it holds for the error as a whole to the extent that the filling does not manufacture a false peak, and that depends on the reconstructor.

4.3.2 Most of the misalignment contrast is hidden share, and persistence adds about a fifth

Three windows are involved (Section 2.1). The climatological window S_c is a median of 12 days from the true window S (90th percentile 29 days, 30 or more days in 9.5% of cases, identical in 4%), so it is a noisy proxy: a gap that covers all of S_c covers all of S in 44% of cases, part of it in 52% and none of it in 4%, while a gap clear of S_c leaves S clear in 94%. The reconstructed window S_r is a median of 0 to 3 days from S but 9 to 14 days from S_c, because the filling has information

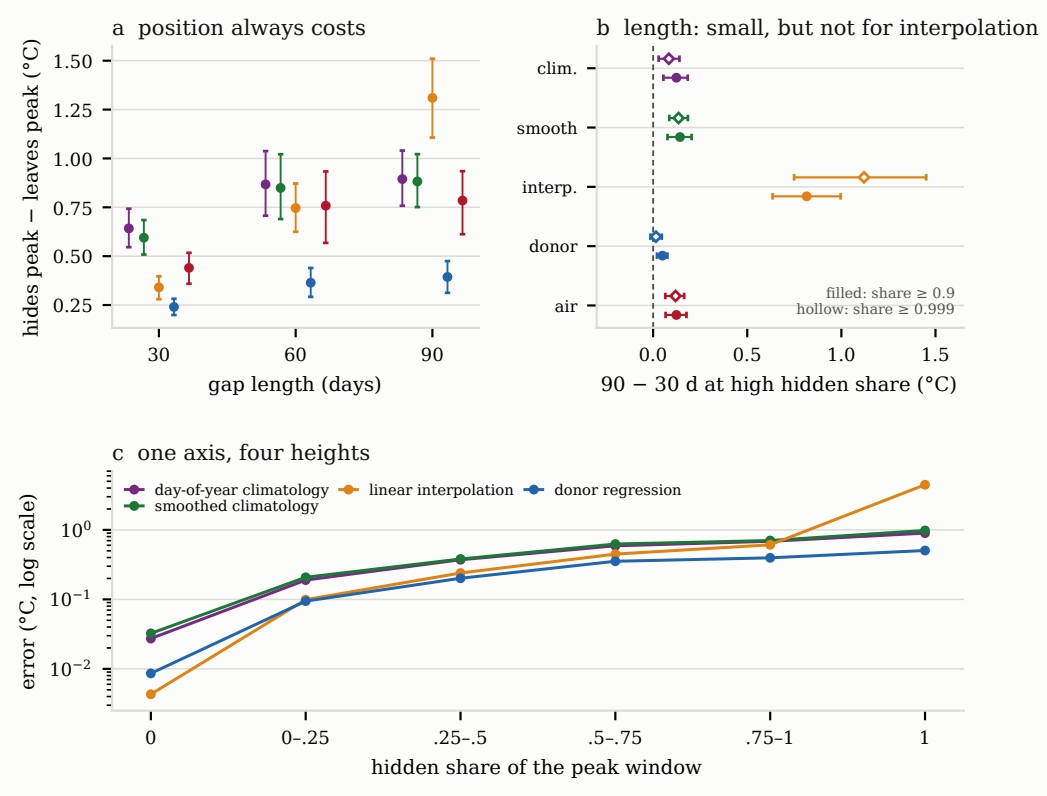


Figure 3. The support effect is universal in sign and conditional in size. (a) Difference in mean absolute error between an outage that covers the whole climatological peak window and one that leaves it clear, within the same station-years, at fixed length (30, 60, 90 days), for five reconstructors. (b) Difference between a 90-day and a 30-day outage within station-years at a true hidden share of at least 0.9 (filled; 48 station-years, 12 networks) and at least 0.999 (hollow; 11 station-years, 6 networks); only linear interpolation has a large length effect. (c) Mean absolute error against the true hidden share of the peak window for the four reconstructors with a series on the external designed panel (a mixed archive; the air-temperature regression has no series there). Panels (a) and (b) use the crossed panel of 1,899 outages and 12 networks with 2,000 network-clustered bootstrap draws; panel (c) uses the 13,017 external outages with group-clustered intervals. Source data: `fig3_source.csv`.

about the year that S_c lacks. When S_r lands away from S the error is larger by 0.66, 0.63, 0.66, 0.32 and 0.49 °C for climatology, smoothing, interpolation, donor and air regression. Most of that contrast is not misplacement. Cases whose reconstructed window moves are overwhelmingly cases whose true window is hidden, and the hidden share is a common cause of both. Held fixed within strata of the climatological class and the true hidden class, the contrast falls to 0.19 [0.07, 0.26], 0.23 [0.11, 0.30], 0.13 [0.07, 0.17], 0.21 [0.14, 0.26] and 0.17 °C [0.07, 0.25], between 35% and 80% lower (Figure 4b). What remains is the false-peak channel. A window switch is also rarely harmless: for climatology 60% of cases switch, 14% of those within 0.1 °C of the truth and a median 0.50 °C away, and the 7% that switch to a window disjoint from S are a median 0.63 °C away. Within the stratum with the true window observed a switch is by identity a false peak, so the contrast there (0.55 °C for climatology) restates the size of the false peak and is not independent evidence for a mechanism.

Residual persistence and the systematic offset are separable by intervention (Figure 4c). The median lag-1 autocorrelation of the residual inside a gap is 0.84 to 0.88 for every method. Permuting the residuals within the gap, which keeps their distribution and mean and destroys the persistence, lowers the mean absolute error by 0.08 °C [0.05, 0.12] for climatology, 12 to 27% depending on the reconstructor. Removing the gap mean while keeping the fluctuations lowers it by 0.24 °C [0.19, 0.30], 49 to 59%, and keeping only the mean leaves three quarters of the error for climatology. Errors therefore accumulate coherently, but the larger part of what a hidden window costs is a systematic offset in the filled days — the climatological and regression fills regress towards the mean, and the year's hottest window is by construction a maximum — and the persistence of what is left contributes about a fifth. A competing thermal peak does not order the error: the relation to the temperature gap to the best disjoint window is non-monotone and flat for donor regression (supporting information).

4.3.3 A model that cannot read the peak-window share keeps as much as one that can

Nothing in this section makes the support rule sufficient, and nothing yet says what a fitted score reads that the rule does not. Figure 4d adds the information to

one gradient-boosted learner in layers and, at each step, reports the retention at the 5% cap under the same nested cross-fit. On the external panel the model that sees only gap geometry and season coverage keeps 63.2% of the candidates; donor information lifts it to 68.7%, reconstruction diagnostics (the reconstructor's own training residual and the disagreement among reconstructors) to 74.0%, and station history adds nothing (73.4%). The peak-window share $\hat{h}$ is the last layer, and with it the model keeps 73.9%. Over three fold seeds the difference from the same model without $\hat{h}$ is +0.4, +0.8 and +0.3 points, every interval includes zero, and on the development panel it is −0.5, +0.4 and +0.5 points (Table 3). The model that cannot read $\hat{h}$ already keeps 6.7 to 7.0 points more than the peak-window rule (9.3 to 10.7 on development), every interval excluding zero. The gain of the fitted score over the transparent rule therefore does not depend on the engineered support feature.

This does not make support unimportant. A model with less to work from needs it: added to gap and season coverage alone, $\hat{h}$ lifts retention by 2.4 to 5.0 points, and added to a small logistic model on six features by 6.2 to 6.8 points (external) and 11.4 to 16.3 (development). Support is a compact and auditable summary of what a flexible model can derive from the gap calendar, season coverage and donor availability. The gain is also concentrated where gaps are consequential: in the stratified outages the model without $\hat{h}$ keeps 61% and the peak-window rule 51%, and in the transplanted real geometries 93% and 91%. Finally, a better support estimate has limited room to help the fitted score. A rule that reads the *true* hidden share would keep 5.0 to 5.9 points more than the climatological rule (8.2 to 9.3 on development), but the generic model given the true share gains only 0.6 to 1.8 points over the one given $\hat{h}$ (2.8 to 4.1 on development). Locating the year's own peak better is worth pursuing for the transparent rule; it is not where the fitted score's advantage comes from.

4.4 What it changes for the decision

4.4.1 Retention at the cap, and the comparison that was pre-specified

Figure 5a gives the retention each rule achieves at the pre-registered 5% cap, averaged over the three fold seeds, on the external and development panels. On the

Table 3. One learner, one training set and one calibration: retention at the 5% cap as information is added, mean of three fold seeds. The change columns give the range over the seeds of the paired difference from the row above (from the fitted score and from the peak-window rule for the last two rows), in points; group-clustered 95% intervals are in the supporting information. The rule rows use no model; the italic rows read the true hidden share and are not deployable.

model or rule	external (%)	development (%)	change, external (points)	change, development (points)
gap-length rule	39.6	30.5	—	—
peak-window rule	66.6	60.4	+27.0	+29.9
gap geometry and season coverage	63.2	58.7	−2.0 to −5.3	−1.2 to −2.1
+ donor information	68.7	62.4	+4.3 to +6.9	+3.2 to +4.9
+ reconstruction diagnostics	74.0	70.2	+4.8 to +5.7	+6.1 to +9.0
+ station history (generic model)	73.4	70.4	−0.7 to −0.2	−0.4 to +0.6
+ peak-window share $\hat{h}$ (fitted score)	73.9	70.5	+0.3 to +0.8	−0.5 to +0.5
certificate bound	72.9	72.7	—	—
generic model + true hidden share	75.0	73.9	+0.6 to +1.8	+2.8 to +4.1
true-hidden-share rule	72.0	69.1	+5.0 to +5.9	+8.2 to +9.3

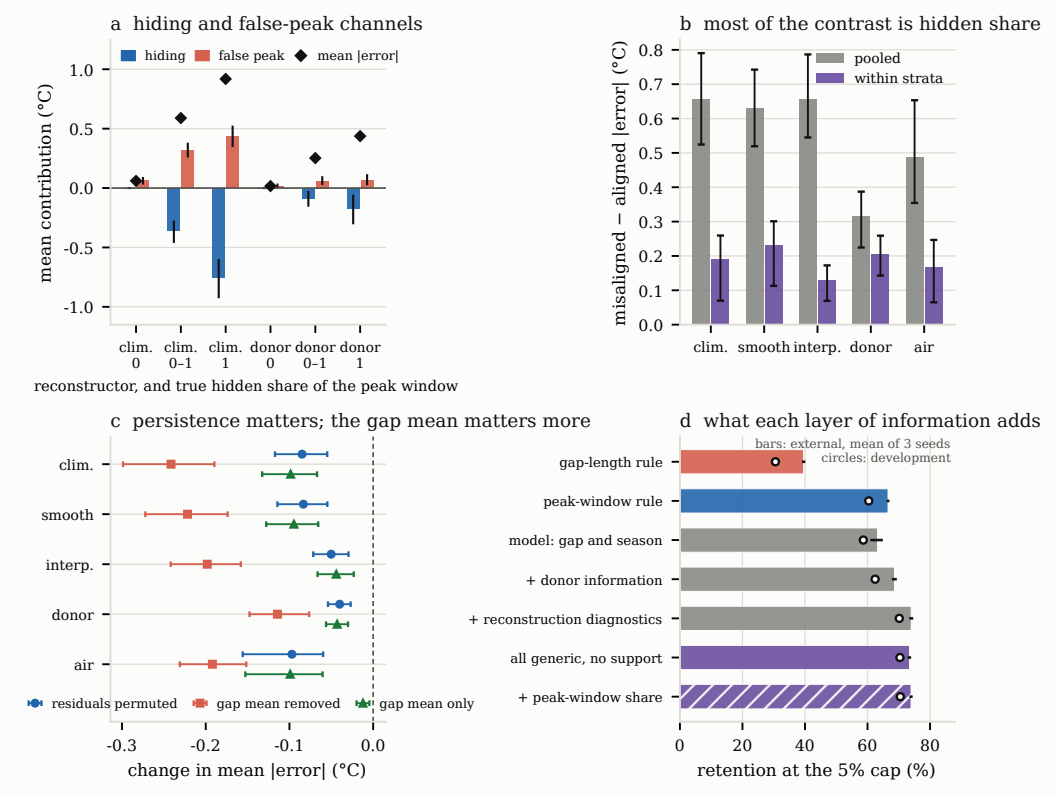


Figure 4. Hiding, false peaks, persistence and the information a fitted score uses. (a) Mean contribution of the hiding channel $\bar{e}_S$ and the false-peak channel F (Equation 3) and mean absolute error, by true hidden share of the peak window (0, between, 1), for the day-of-year climatology and donor-regression reconstructions. (b) Misaligned minus aligned mean absolute error, pooled and within strata of the climatological class and the true hidden class, for five reconstructors. (c) Change in mean absolute error when the gap residuals are permuted, their mean is removed, or only their mean is kept. Panels (a)–(c) use the 1,899-outage crossed panel (211 station-years, 12 networks) with network-clustered 95% intervals from 2,000 draws. (d) Retention at the 5% cap on the external panel (bars, with the range over three fold seeds) and the development panel (circles) as information is added to one learner; hatching marks the fitted score. Source data: `fig4_source.csv`.

external panel the gap-length rule keeps 39.6% of candidates, the peak-window rule 66.6%, the certificate bound 72.9% and the fitted risk score 74.0%. The fitted-score gain over the peak-window rule is 8.0, 7.3 and 7.0 points at seeds 101, 202 and 303 (Figure 5d). Every point estimate exceeds the five-point margin that was fixed before the comparison and every interval excludes zero, but the lower limits are 5.0, 4.6 and 4.3 points, so we do not claim a gain of at least five points at 95% confidence at every seed. The three seeds re-split one panel, so they are repeated cross-fits and not three confirmations. The development panel reproduces the sign and the size: 70.4% against 60.4%, a gain of 9.7 to 10.3 points. Against the certificate bound the fitted score is not reliably superior: it is +1.6 points [+0.2, +3.1] on the external panel at seed 101 but +1.1 and +0.6 with intervals that include zero at the other two seeds, and −2.6, −1.7 and −2.5 points on development, where the interval includes zero at all three. We therefore report an advantage over a transparent support rule and no reliable advantage over the calibrated bound.

At a fixed 70% retention the ordering is the same and the scale is easier to read: the fitted score admits 2.2% failures on the external panel against 3.7% for the peak-window rule, 8.3% for the gap-length rule and 2.8% for the bound. The gap-length rule is not a straw baseline that a weak comparison flatters: at 70% retention it is the worst of the four by more than five points, which is the cost of asking the wrong question of the archive.

4.4.2 Refusing a record costs information, and hot years are refused like all records

At a 20 °C hot-year threshold on the external panel, 86% of candidates are true hot years. Reconstructing everything refuses none of them but misjudges 1.3% as cool. The fitted score refuses 25% of the true hot years, the certificate bound 26%, the peak-window rule 33% and the gap-length rule 60% (Figure 5c). Those numbers are the retention of the rule, not a property of hot years: a rule that ignored the temperature and kept the same share of records would refuse the same share of hot years. The fitted score refuses 25.5% of all records, so hot years are refused *less* often than records in general, by 1.0 point [0.3, 1.6] at seed 101 and 0.8 to 1.0 points at the other seeds, with the certificate bound at 1.0 and the peak-window rule at 0.1 to 0.3 points, an interval that includes zero. On the development panel the excess

refusal of the fitted score is −0.1, +0.3 and +0.4 points, all intervals including zero. We find no evidence that a screen is selective against hot years; what it costs is the information in the records it refuses. The admitted hot-year share rises by 1.1 points under the fitted score, for a total deviation of +0.7 points with a 95% interval of −0.3 to +1.6, so the archive's composition is not distorted much either.

Whether that information is worth giving up depends on what a refused record costs, which the error rate among admitted records does not say. Judging each admitted year from its reconstruction and treating a refusal as an abstention, the fitted score is wrong on 0.45% of candidates against 1.6% when everything is reconstructed, at the price of abstaining on 25%. It pays only if one abstention costs less than 0.043 to 0.046 of a wrong hot-year judgement (0.062 to 0.063 on the development panel); if a missed hot year is five times as costly as a false alarm the break-even is 0.15 to 0.16, and at ten times 0.28 to 0.30. A review that costs a tenth of a wrong judgement would therefore make the fitted score worse than reconstructing everything under equal costs, and better only if misses are weighted heavily. The gap-length rule, which abstains on 60%, breaks even at 0.024.

4.4.3 Most real networks need no screen, and the gap calendar says which

Figure 5b opens the pooled result of the 418 real calendars. Pooled, admitting every reconstruction puts 2.6–4.1% of records above the tolerance, under the cap at every point estimate, although the clustered intervals are wide (0.3 to 7.8% when only the target station fails). The screens keep 93–96% of candidates and reduce the failure rate to 0.5–1.7%. The fitted-score advantage over the peak-window rule shrinks to 0.2–2.4 points, and only 2 of 9 level-by-seed intervals exclude zero. Where the peak window is usually observed, screening has almost nothing to do: judging hot years from the unscreened reconstruction is wrong in one of 418 groups.

The pooled rate hides where the failures are. Two networks exceed the cap when only the target station fails (Czech Morava, 67% on six station-years; Norwegian basin 26, 50% on three) and four when every donor fails; the other twelve networks, or ten at the all-donor level, sit at 0 to 3.7% (0 to 2.1%). The two failing networks are the ones whose gaps cover the peak window: the share of records

hiding more than half of the climatological window is 100% and 50% there against 0 to 22% elsewhere, and the median gap is 364 and 48 days against 2 to 13 days in ten of the other twelve. The composition rule of Section 3.5 recovers this without a real label. Built from designed outages only, it predicts a pooled failure rate of 3.5% against 2.6–4.1% observed, and it never says "no screen" for a network that exceeded the cap (zero of two to four), at the price of advising a screen for one to three of the others; it is right for 11 to 13 of 14 networks (exact 95% interval for the proportion 49 to 99.8%). Among the designed archives themselves, composition tracks the between-archive variation only modestly (Spearman 0.35 on the 110 development networks and 0.61 on the 41 external groups; mean absolute error 5.8 against 6.6 points for the pooled rate on development), and 90% and 98% of those archives exceed the cap, so the rule advises a screen for all of them. The real evidence rests on two to four networks that need a screen and on 3–6 station-years in each, so the rule is a description of this replay, not a validated decision boundary: it says what to look at — the share of records whose gap covers the peak window — and it has not been shown to hold for an archive from another agency.

4.5 The risk rule delivers the risk, and the earlier number did not

Two properties of the selector need to be separated because an earlier version of this analysis conflated them. The first is the population risk of the threshold the rule selects: if the archive were infinite, would the admitted set violate the 5% cap? The second is the error rate observed in a finite fresh batch, which carries its own sampling noise. We re-ran the selector on 2,000 synthetic clustered archives per scenario, keeping the deployed rule identical to the one used on the river panels, and measured both.

With the deployed simultaneous bound, the selected threshold's *population* failure rate exceeds 5% in 0.4% of repetitions in the baseline scenario, 0.6% near the cap, 0.4% with unequal group sizes and 2.4% with only twelve calibration groups. The finite fresh batch exceeds 5% far more often — 7.8%, 7.5%, 4.6% and 12.2% in the same scenarios — because the batch adds noise that the rule cannot control. The 8.5% figure quoted in the previous version of this work was the second quantity, and reading it as the selector's error rate was a misinterpretation; the rule is con-

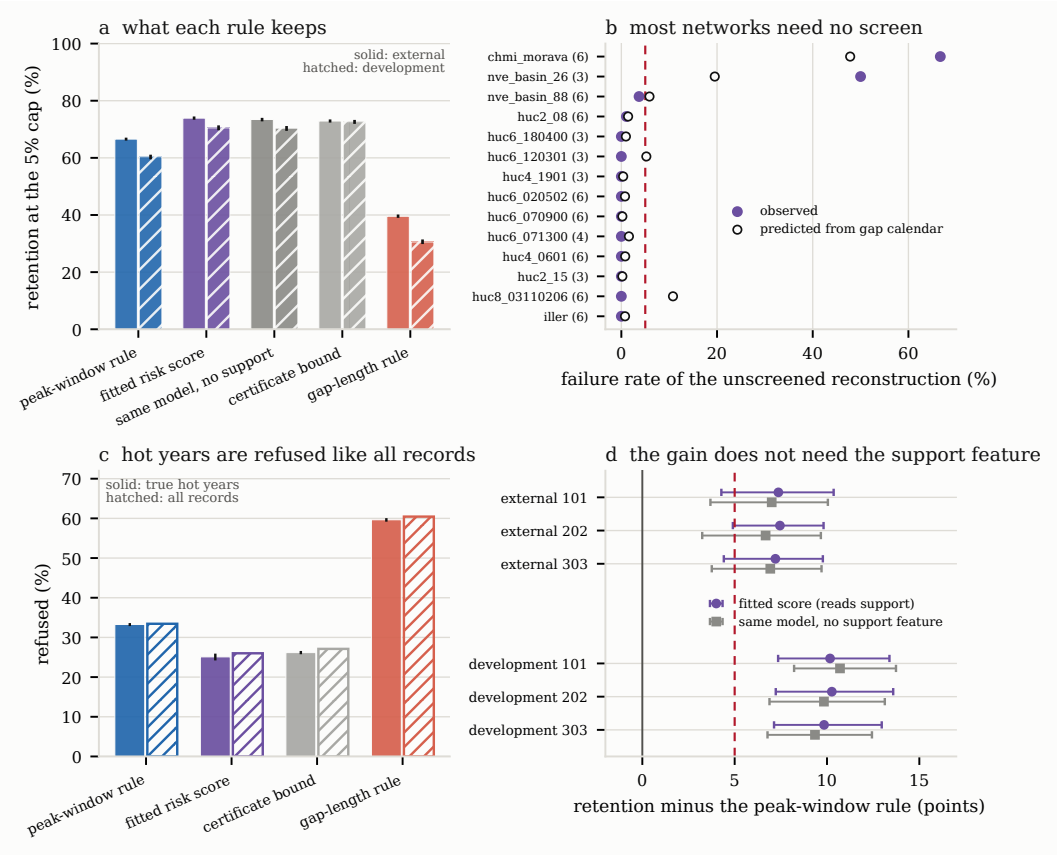


Figure 5. What each rule keeps, and what the keeping costs. (a) Retention at the 5% cap of the admitted reconstruction's error, by rule, on the external panel (solid) and the development panel (hatched); bars are the mean of three fold seeds and the line spans the seeds. The reference arm that admits every reconstruction does not satisfy the cap and is omitted. "Same model, no support" is the fitted score without the peak-window share. (b) Failure rate of the unscreened reconstruction in each of the 14 real-calendar networks when only the target station fails (filled), and the rate predicted from the network's gap calendar by a composition rule built on designed outages only (open); labels give the number of target station-years; the dashed line is the 5% cap. (c) At a 20 °C hot-year threshold on the external panel, the share of true hot years refused by each rule (solid) against the share of all records it refuses (hatched), which is what a rule blind to temperature would refuse of the hot years; the solid bar's line spans the three fold seeds. (d) Retention minus the peak-window rule, with group-clustered 95% intervals, by fold seed, for the fitted score and for the same model without the peak-window share; the dashed line is the pre-specified five-point margin. Source data: `fig5_source.csv`.

servative, not miscalibrated. The price is retention: the simultaneous bound keeps 18.5% of synthetic candidates where a pointwise bound keeps 35.9%.

The extreme percentile the rule reads, $1 - 0.05/86 \approx 0.9994$, is numerically delicate at 1,000 bootstrap draws. Re-running the selection with 10,000 draws changes the chosen threshold in 42 of 150 fold-by-score cells. The decision consequences are small for the score this paper recommends — the fitted score's retention moves by at most 1.0 point — and larger for the discrete rules, where one grid step can move 3.9 points of retention. We therefore describe the selected threshold as a calibration decision at this granularity, not as a stable optimum.

4.6 What the repair changed, and what it did not

The previous version of this analysis declared the policy reconstruction as the admitted object but calibrated the failure event on a different reconstruction. On the external panel the two failure definitions disagree on 898 of 13,017 cases — 6.9% — and the mismatch mattered: with the event measured on the certificate, the fitted score's gain over the peak-window rule on the external panel was 2.7 points; measured on the admitted reconstruction it is 7.2 points under the same tie rule. Most of the change (about 4.5 points) comes from measuring the right object, and about 1.3 points from admitting whole tie blocks instead of jittering the scores. Neither repair was optional: a rule whose stated object is the policy reconstruction cannot be calibrated on a different one, and a rule whose calibration admits half of a tie block and whose deployment admits all of it is two rules.

5 Discussion

5.1 What is new relative to gap-length guidance

The standing guidance is not wrong about what it was derived from. On the estimands for which it was derived — annual means and annual signal parameters that integrate over the year — we reproduce its behaviour: the error grows with the amount of record removed and the position of the removal is second order. Our disagreement is about the transfer of that result to a named extreme. For the hottest 30-day mean the ordering inverts, and the inversion is not small: at the same gap length, a badly placed 30-day outage costs about twenty times a well-placed one,

while a 180-day outage that leaves the peak window observed costs almost nothing. A completeness rule cannot see this because the quantity that matters — the share of the statistic's support that is absent — is not a function of the number of missing days alone. Earlier work already tested timing (Johnson et al., 2021) and admits partial-season records on the climatological position of the hottest window (Beaufort et al., 2022), and selective prediction already withholds a prediction when its predicted risk is high (Geifman & El-Yaniv, 2017; Denis et al., 2020); the contribution here is the attribution that those studies did not make: a controlled position and length contrast within station-years, an exact split of the error into a hiding and a false-peak channel, a measurement of how much of a fitted screen's gain depends on the support feature, and the cost of refusal against a hot-year-blind null.

This also changes what a recovery experiment has to control. A comparison across real gaps cannot identify the support effect, because length and support are entangled in exactly the records where the decision is made, and the pooled designed panels are no better: length and the true hidden share still correlate at +0.52 in them. The crossed panel is what makes the claim testable, and the claim it supports is conditional: the support share is the right conditioning variable for every reconstructor we tested, but the coefficient on it, and whether a residual length effect survives, are properties of the reconstructor rather than of rivers.

5.2 Why the fitted score adds seven to ten points, and what it does not need

The gain over the peak-window rule is real and large enough to matter for an archive, but the ladder in Table 3 shows where it comes from. Donor information and the reconstructor's own diagnostics — how well neighbouring stations track the target, how large the training residual was, how much the five reconstructions disagree — supply the whole of it. The peak-window share adds nothing once those are present. The score does read something that the support share does not contain, and the ladder says what: information about the reconstruction, not about the target statistic's support. The fitted score is therefore not a more elaborate version of the support rule but a different instrument that agrees with it, and an agency that cannot train one gives up 7 to 8 points of retention in exchange for a rule it can audit.

Neither place justifies a general claim of superiority, and the comparison with the certificate bound says so directly. On the development panel the bound keeps more records at the same risk; on the external panel the difference is at most 1.6 points and crosses zero at two of the three seeds. The honest reading is that a fitted score is one way to recover the information a support rule leaves on the table, not the only way and not demonstrably the best way. What the support rule does better than either is explain itself: its decision is a statement about the record, and an agency can audit it without the model. That is also the sense in which support is the right variable: it is the best single observable, it is what a small model needs to be competitive, and it supplies the vocabulary for the exact decomposition above, though it is not what a flexible model needs.

5.3 The regime boundary: when screening cannot pay

The real-calendar result is the one that should govern adoption, and it has two parts. The gaps that monitoring programmes actually leave are short and mostly off-peak: a median of seven days, a median of 2.2% of the year missing, and 5.7% of records hiding more than half of the peak window. In that regime the unscreened archive sits under the cap at the point estimate, every screen removes only a small tail, and the largest difference between the fitted score and the transparent rule is 2.4 points. But the pooled rate averages over networks that behave very differently: two to four of the 14 networks fail at 7 to 83% while the rest are essentially clean. A decision about whether to screen is therefore a decision about records, or about the networks whose gaps cover the peak window, and not about the pooled calendar; the share of records hiding more than half of the climatological window separated the failing networks from the others in this replay, but the evidence is two to four networks resting on 3 to 6 station-years each, and we do not offer it as a decision boundary.

Screening earns its cost in the other regime: long gaps, at or over the peak window, which is where the designed panel lives, the fitted score's advantage over the support rule is 7 to 8 points, and the support rule's advantage over the length rule is about 27. The operational content of the paper is therefore: compute the share of the peak window each record hides before looking at any label; screen the

records for which it is large; and judge the cost of refusing a record against the cost of a wrong judgement before screening more.

The hot-year result makes the last point. A screen that lowers the failure rate among admitted records while refusing 25 to 60% of the true hot years looks costly, but the refusal is the screen's retention, not a selection against hot years: hot years are refused no more often than records in general. What a screen spends is information, and whether that is worth spending depends on a cost the data cannot supply. Under equal costs a screen pays only if a review or a re-run of a refused record costs under about a twentieth of a wrong judgement. The two quantities — the error among admitted records and the abstention rate — have to be reported together; one without the other can always be made to look good.

5.4 Limitations

The external panel is already opened. Every comparison on it is a local cross-fit: the model and the threshold for a held-out group are learned from labelled outages on the other groups of the same panel. That answers whether an agency that holds complete years and labelled outages for part of its network can calibrate a screen for the rest. It does not answer whether a model and thresholds frozen on the development archive can be carried unchanged to another agency; that needs a sealed panel, and the three fold seeds re-split one panel and are not three confirmations. The development panel is a sanity replication rather than an independent test. The real-calendar replay is a development-period probe: 418 groups from 59 source gaps on 67 target station-years in 14 networks, with two to four networks that need a screen, so its intervals are wide and its point estimates should not be extrapolated to a national archive or to an unusually hot or low-flow year.

The mechanism analysis rests on one controlled panel: 211 station-years from 24 stations in 12 development networks, one outage per cell, five reconstructors fitted leave-one-year-out. The sign of the hiding channel, a cool bias in the hidden window, is a property of fills that regress towards the mean and may differ for a reconstructor that does not. The interventions on the residuals are numerical experiments, not natural-process causal identification, and the permutation keeps the residual distribution but not its dependence on the weather of the year. The fitted

score's advantage over the peak-window rule is concentrated in the stratified outages, which are single runs, where run count and total missing days identify the design arm; in the transplanted real geometries the two differ by two points (93% against 91%).

The cell edges of the composition rule (gaps of 14, 30 and 90 days; overlap of 0 and 0.5) were fixed by us after we had seen the replay's per-network summaries and before the rule was run; no alternative was tried, but the replay is not a blind test of them.

The tolerance of 0.5 °C is a judgement, not a measurement. It corresponds to about half the within-station interannual standard deviation of the hottest 30-day mean in this panel, but an agency with a different use — a criterion exceedance, say — would set a different one, and the retention numbers scale with it. The costs in the abstention analysis are scenarios, not measurements; we report the break-even cost rather than assume one.

The record's measurement basis is not uniform. About 74% of development cases are provider-declared daily means, 15% are project-aggregated sub-daily data, 8% have an unspecified daily definition, and 2% are morning reference readings; on the external panel about 94% are declared daily means. We re-selected the thresholds inside the declared-daily-mean subset and the ranking does not depend on the mixed-measurement providers: the fitted score's advantage over the peak-window rule is 8.0, 7.3 and 7.3 points on the external subset and 8.8, 8.7 and 10.4 on the development subset, close to the full-panel values, and its comparison with the certificate bound remains interval-crossing.

Finally, the support share is defined against the year's own peak window, which is unknown at decision time. The deployable rule uses the climatological window, which is a median of 12 days from the true one. A rule that read the true share would keep 5.0 to 5.9 points more than the climatological rule on the external panel, but would add at most 1.8 points to the fitted score, so predicting the year's window from antecedent conditions is worth building for a transparent rule and we have not built it.

6 Conclusions

A completeness rule assumes that the amount of a record that is missing decides whether the record can be used. We tested that assumption by separating what real records entangle. The 5,114 complete station-years carry 53,721 designed outages from two designs that compare cases across a mixed archive; a further 1,899 outages on 211 station-years cross gap length and position within the same year; and 418 real monitoring calendars were replayed from 59 recorded gaps.

The share of the target statistic's own support that a gap hides orders the error of an annual thermal extreme, whereas the amount missing orders the error of an annual mean; the two orderings are opposite, and within station-years a 30-day gap that covers the climatological peak window costs 0.64 °C more than the same gap clear of it. The position effect ranges from 0.24 to 0.64 °C at 30 days across five reconstructors; a length effect remains but is five to eight times smaller, except for linear interpolation. The error splits exactly into a hiding channel, a cool offset in the filled days, and a false-peak channel; most of the apparent cost of a reconstructed window that lands elsewhere is hidden share, and residual persistence accounts for about a fifth of the error.

For the admission decision, conditioning on support keeps 67% of an external panel at a 5% error cap where a gap-length rule keeps 40%, and a fitted risk score keeps 74%, 7 to 8 points more than the transparent support rule and not reliably more than a calibrated certificate bound. The fitted score does not need the support share to do so: the same model without it keeps 73%, and the gain comes from donor and reconstruction diagnostics. On the real calendars the unscreened archive sits under the cap at the point estimate, but two to four of 14 networks fail badly, and screens refuse hot years in proportion to all records, so what they spend is information, not hot years. Support is the right conditioning variable for a transparent rule and for a small model; a flexible model already derives it. Its value is confined to archives whose gaps are consequential, and inside that regime a transparent support rule captures most of what a complex one does.

Open Research Statement

All data are public monitoring records reused under their providers' terms, and the derived tables are regenerated by the scripts listed below. The repository holds the panel-building code, the analysis scripts, the frozen thresholds and the figure sources; the schematic figures are draw.io sources under `figures/wrr/drawio/`, exported to PDF with draw.io 32.5.5, and every data figure is a Python script writing the PDF that appears here, with the table behind each panel in `results/wrr_submission_ready/major_revision_20261020/p0_06_figures/` and `results/wrr_submission_ready/major_revision_20261020b/r2_06_figures/`. The scripts that produce every number in this manuscript are `scripts/330` to `scripts/336` (aligned admission) and `scripts/340` to `scripts/345` (design audit, same-capacity ladder, three-window decomposition, hot-year null, screening-need rule and the two figures that depend on them); `scripts/337` exports the schematics and `scripts/338` re-checks the bibliography, with the earlier evidence chain in `scripts/301` to `scripts/324`. Inputs, output hashes and software versions are recorded in each run's `protocol_fingerprint.json`. A permanent archive with a DOI is planned at acceptance; the current repository is a development repository and is not offered as the archival copy. Provider licences and the processing steps that produce the daily series are documented in `DATA_RIGHTS.md` and the supporting information.

Acknowledgments

We thank the national monitoring agencies listed in the supporting information for maintaining and publishing the records this study reuses. [Funding and author contributions pending author confirmation.]

References

Angelopoulos, A. N., Bates, S., Fisch, A., Lei, L., & Schuster, T. (2022). *Conformal risk control.* arXiv preprint arXiv:2208.02814. doi: 10.48550/arXiv.2208.02814

Bai, S., Fang, Z., & Chen, J. (2026). Risk-controlling predictive sets for time-series events under selective observation with finite-sample guarantees. *Axioms*, *15*(9), 706. doi: 10.3390/axioms15090706

Beaufort, A., Diamond, J. S., Sauquet, E., & Moatar, F. (2022). Spatial extrapolation of stream thermal peaks using heterogeneous time series at a national scale. *Hy-*

drology and Earth System Sciences, *26*(13), 3477–3495. doi: 10.5194/hess-26-3477-2022

Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, *51*(8), 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x

Corona, C. R., & Hogue, T. S. (2025). Machine learning in stream and river water temperature modeling: a review and metrics for evaluation. *Hydrology and Earth System Sciences*, *29*(12), 2521–2549. doi: 10.5194/hess-29-2521-2025

Denis, C., Hebiri, M., & Zaoui, A. (2020). Regression with reject option and application to kNN. In *Advances in neural information processing systems* (Vol. 33).

Falcone, J. A. (2011). *GAGES-II: Geospatial attributes of gages for evaluating streamflow* (Tech. Rep.). U.S. Geological Survey. doi: 10.3133/70046617

Geifman, Y., & El-Yaniv, R. (2017). Selective classification for deep neural networks. In *Advances in neural information processing systems* (Vol. 30).

Grey, V., Hatt, B. E., Fletcher, T. D., Smith-Miles, K., & Coleman, R. A. (2024). Analysis of irregularly sampled stream temperature time series: challenges and solutions. *Journal of Hydrology*, *636*, 131244. doi: 10.1016/j.jhydrol.2024.131244

Hare, D. K., Benz, S. A., Kurylyk, B. L., Johnson, Z. C., Terry, N. C., & Helton, A. M. (2023). Paired air and stream temperature analysis (PASTA) to evaluate groundwater influence on streams. *Water Resources Research*, *59*(4), e2022WR033912. doi: 10.1029/2022WR033912

Huang, F., Qian, B., & Ochoa, C. G. (2023). Long-term river water temperature reconstruction and investigation: a case study of the Dongting Lake Basin, China. *Journal of Hydrology*, *616*, 128857. doi: 10.1016/j.jhydrol.2022.128857

Isaak, D. J., Wenger, S. J., Peterson, E. E., Ver Hoef, J. M., Nagel, D. E., Luce, C. H., ... Parkes-Payne, S. (2017). The NorWeST summer stream temperature model and scenarios for the western U.S.: a crowd-sourced database and new geospatial tools foster a user community and predict broad climate warming of rivers and streams. *Water Resources Research*, *53*(11), 9181–9205. doi: 10.1002/2017WR020969

Jackson, F. L., Fryer, R. J., Stirling, D., & Malcolm, I. A. (2025). The influence of equipment bias on reported temperature trends: implications for river temperature monitoring networks. *River Research and Applications*, *41*(8), 1604–1615.

doi: 10.1002/rra.70012

Johnson, Z. C., Johnson, B. G., Briggs, M. A., Snyder, C. D., Hitt, N. P., & Devine, W. D. (2021). Heed the data gap: guidelines for using incomplete datasets in annual stream temperature analyses. *Ecological Indicators*, *122*, 107229. doi: 10.1016/j.ecolind.2020.107229

Kratzert, F., Nearing, G., Addor, N., Erickson, T., Gauch, M., Gilon, O., ... Matias, Y. (2023). Caravan — a global community dataset for large-sample hydrology. *Scientific Data*, *10*, 61. doi: 10.1038/s41597-023-01975-w

Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., & Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, *113*(523), 1094–1111. doi: 10.1080/01621459.2017.1307116

Li, H., Deng, X., & Smith, E. (2017). Missing data imputation for paired stream and air temperature sensor data. *Environmetrics*, *28*(1), e2426. doi: 10.1002/env.2426

Mauger, S., Shaftel, R., Leppi, J. C., & Rinella, D. J. (2017). Summer temperature regimes in southcentral Alaska streams: watershed drivers of variation and potential implications for Pacific salmon. *Canadian Journal of Fisheries and Aquatic Sciences*, *74*(5), 702–715. doi: 10.1139/cjfas-2016-0076

Mauger, S., Shaftel, R., Trammell, E. J., Geist, M., & Bogan, D. (2015). Stream temperature data collection standards for Alaska: minimum standards to generate data useful for regional-scale analyses. *Journal of Hydrology: Regional Studies*, *4*, 431–438. doi: 10.1016/j.ejrh.2015.07.008

Mohseni, O., Stefan, H. G., & Erickson, T. R. (1998). A nonlinear regression model for weekly stream temperatures. *Water Resources Research*, *34*(10), 2685–2692. doi: 10.1029/98WR01877

NASA Langley Research Center. (2026). *POWER project data sets: daily 2 m air temperature (MERRA-2).* NASA Prediction Of Worldwide Energy Resources, NASA Langley Research Center. (Accessed 2026. `https://power.larc.nasa.gov`)

Rubin, D. B. (1976). Inference and missing data. *Biometrika*, *63*(3), 581–592. doi: 10.1093/biomet/63.3.581

Sanikhani, H., Khorsandi, M., Déry, S. J., & St-Hilaire, A. (2026). Large-scale climate

teleconnections and multiscale controls on river water temperature. *Journal of Hydrology*, *679*, 136301. doi: 10.1016/j.jhydrol.2026.136301

Wang, S.-h. (2025). Long-term reconstruction and variation characteristics of water temperature in mainstream Yangtze River. *Journal of Changjiang River Scientific Research Institute*, *42*(10), 54–63. doi: 10.11988/ckyyb.20240856

Washington State Department of Ecology. (2024). *Water quality standards for surface waters of the state of Washington, WAC 173-201A-020 and WAC 173-201A-200.* Washington Administrative Code.

You, Y., Wang, Y., Tao, J., Zhao, L., Wang, S., Zhang, Y., ... Wang, D. (2026). Unveiling river thermal regimes in the Yangtze river basin, China, with a hybrid deep learning model. *Journal of Environmental Management*, *398*, 128460. doi: 10.1016/j.jenvman.2025.128460

Zhang, X., Alexander, L., Hegerl, G. C., Jones, P., Klein Tank, A., Peterson, T. C., ... Zwiers, F. W. (2011). Indices for monitoring changes in extremes based on daily temperature and precipitation data. *WIREs Climate Change*, *2*(6), 851–870. doi: 10.1002/wcc.147